

CLAUDE.md V3 - Streamlit Edition

Project Vision

Build a full end-to-end machine learning product in pure Python that helps users determine whether an appliance is fairly priced and discover better-value alternatives.

Business Problem

Consumers often do not know whether appliance prices are justified. The system estimates fair value, computes value scores, and recommends better alternatives.

Technology Stack

Python 3.11+, Pandas, NumPy, Scikit-Learn, XGBoost, SHAP, Requests, BeautifulSoup, Matplotlib, Plotly, Streamlit, Joblib.

Project Structure

appliance-pricing-platform/ data/raw data/processed notebooks/ src/ models/ app.py
requirements.txt README.md CLAUDE.md

Data Requirements

Source: Smartprix Fields: Product Name Brand Category Price Rating Review Count Energy Rating Capacity Warranty Technical Specifications

Models

Linear Regression Ridge Regression Lasso Regression Decision Tree Regressor Random Forest Regressor XGBoost Regressor SHAP Explainability

Workflow

1. Scrape Smartprix. 2. Clean dataset. 3. Perform EDA. 4. Feature engineering. 5. Train baseline Linear Regression. 6. Train advanced models. 7. Compare metrics. 8. Save best model. 9. Build pricing intelligence engine. 10. Build alternative recommendation engine. 11. Add SHAP explainability. 12. Build Streamlit UI. 13. Deploy to Streamlit Cloud.

Streamlit Pages

Page 1: Appliance Analyzer Inputs: Brand Category Capacity Energy Rating Warranty Actual Price Outputs: Predicted Fair Value Value Score Verdict Page 2: Alternative Recommendations Top 5 similar products ranked by value. Page 3: Why This Price? SHAP explanations.

Milestones

P0: Scraping Cleaning EDA Linear Regression P1: Ridge Lasso Random Forest XGBoost P2: SHAP Recommendation Engine P3: Streamlit Deployment